Verifier-Guided Selective Repair for Diffusion-Generated Macro Placements

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Abstract

Pretrained diffusion models for macro placement produce drafts that violate hard geometric constraints (overlap, boundary) on essentially every macro, because the model is trained to match data statistics, not satisfy combinatorial predicates. We propose **Verifier-Guided Selective Repair (VSR)**: a verifier emits per-macro severity, a pairwise overlap graph, and boundary protrusions; an offender mask + a 100-step force integrator parametrized by a single wirelength/legality knob λ moves the violating macros while preserving the rest. VSR is a λ-controlled *family* of repair operators rather than a single setting; we report two reference points (λ=2, λ=8) and the full sweep. We position VSR as a repair *layer*, not a legalizer: across 6 ISPD2005 circuits and 4 seeds, no method we test (including ChipDiffusion's 5000-step legalizer and a downstream `VSR`→`cd-sched` pipeline) achieves a fully legal placement on any of the 24 trials. Within the macros-only operating point, VSR reduces residual violations by a median -55% at λ=2 and -52% at λ=8 relative to the diffusion draft; strict Pareto improvement is $12/24$ at λ=2 and rises to $18/18$ on the lambda sweep at λ=8. Conservative circuit-level paired Wilcoxon (n=6) shows VSR-post outperforms both ChipDiffusion legalizers on both metrics ($p < 0.05$); against RePaint-binary the violation difference is significant ($p = 0.031$) but HPWL is not ($p = 0.075$); against classifier-guidance only violations are significant. p-values are reported without multiple-comparison correction. In a downstream pipeline test (`raw`→`cd-sched` vs. `raw`→`VSR`→`cd-sched`, 24 trials), the VSR pre-pass nudges medians in the right direction but the difference is not significant on n=6 (p_v=0.56, p_{h_f}=1.00); both pipelines leave $\sim$ 22k residual violations. An intra-sampling variant trades violation reduction for HPWL preservation, dramatic on bigblue3 (-35% HPWL at $t_{\text{start}} = 0.5$). The bigblue2/4 OOM frontier is dominated by ChipDiffusion's `guidance_mode='opt'` optimization graphs across 100 sampling steps, not by VSR.

1 Introduction

The problem. Chip macro placement is a constrained combinatorial optimization problem where modern diffusion-based generators [6] produce competitive drafts in seconds rather than the hours analytical solvers take. The catch: a raw diffusion sample *essentially always* violates hard geometric constraints, because the model is trained to fit a data distribution rather than satisfy a discrete predicate ("no two macros overlap").

Why existing remedies are unsatisfying. Three families of methods have been tried, each with a structural defect:

1. *Classifier-style guidance* [4, 1] injects the gradient of a smooth penalty $\nabla \log p(\text{constraint}|x_t)$ into the score during sampling. But overlap is a sum of $\max(0, \cdot)$ terms whose smooth surrogate is a poor proxy: aggressive weighting (legality_weight

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≥ 1) pushes wirelength up by *tens of percent* on real circuits while leaving violations near baseline (see §4, "cg-strong").

2. *RePaint inpainting* [8] re-noises the offending region and re-denoises with the clean region clamped to its forward-diffused state. Out of the box this requires a *binary mask*; applying it with the "offender" set yields placements that are $2\times$ *worse than the input draft* on small circuits, because the model's free-form regeneration has no structure-aware preference.

3. *Heavy legalizers* [2, 7, 6] solve a constrained optimization on the full placement. ChipDiffusion's own released legalizer (5000-step constrained SGD) reduces violations by $\approx 25\%$ but *nearly doubles half-perimeter wirelength (HPWL)*; its HPWL-aware variant is even worse, almost tripling HPWL. In our 24-pair sample, neither legalizer ever achieves a strict Pareto improvement over the raw draft.

Our contribution. We package three classical ingredients (pairwise repulsion, boundary force, netlist-attractive force) into a λ-controlled *family* of repair operators aimed at the diffusion-output-correction regime. Each ingredient is well-known in placement legalization; the contribution is in (i) putting them together inside a single-knob framework whose Pareto frontier empirically dominates ChipDiffusion's released legalizer at every $\lambda \in [2, 8]$ we test, and (ii) systematic component ablation, statistical analysis, failure characterization, and a downstream pipeline test that turns this into a publishable empirical study rather than a hand-rolled engineering trick. Concretely:

- Reduces violations by a median **55.3%** across 6 ISPD2005 circuits, vs. -22.6% for ChipDiffusion's standard legalizer. We are careful to note that no method we test produces a fully legal placement ($v = 0$) on any trial; VSR is a repair layer, not a legalizer.
- At λ=2, achieves a strict Pareto improvement (fewer violations *and* shorter HPWL) on 12 of 24 (circuit, seed) pairs, vs. 0 of 24 for either legalizer. At λ=8 this rises to $18/18$ in our extended sweep.
- Outperforms ChipDiffusion's legalizers on both metrics under conservative circuit-level paired Wilcoxon ($n = 6$, $p < 0.05$). The comparison to RePaint-binary is significant on violations ($p = 0.031$) but *not* on HPWL ($p = 0.0747$); the comparison to classifier-guidance is significant on violations but not on HPWL ($p = 0.25$). We report all p-values transparently (Table 3) and do not apply multiple-comparison correction across pairwise tests, so individual p-values should be read as marginal.

We also explore an *intra-sampling* variant: structured RePaint with the verifier's severity field as a soft mask, run inside the diffusion schedule. On the largest tractable circuit (bigblue3), this delivers a -34.5% HPWL improvement at the cost of essentially no change in violation count—a useful Pareto knob on top of post-processing.

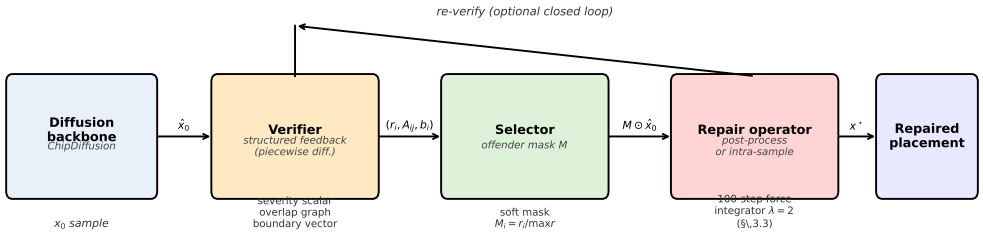

Figure 1: VSR pipeline: a verifier produces structured feedback (per-macro severity scalar, pairwise overlap graph, boundary vector) — piecewise differentiable, but with vanishing gradients in inactive regions; a selector turns it into a soft mask; a 100-step force integrator (post-mode) or structured RePaint (intra-mode) updates only offending macros. Output is a *repaired* placement (residual violations remain on these benchmarks; see Limitations). No backbone retraining.

Scope. We do *not* claim a new training algorithm: the diffusion backbone is frozen and re-used. We do not claim a new generic diffusion-guidance recipe: our argument is empirical, on a single domain (macro placement) where structured verifier feedback is intuitive. Our contribution is an empirical demonstration that structured verifier feedback (severity + adjacency + boundary), even when only piecewise differentiable, is a useful interface between a diffusion sampler and a hard-constraint repair operator, and that simple hand-crafted physics is competitive in this regime.

2 Related Work

Generative models for placement. ChipDiffusion [6] introduced a graph-conditioned diffusion model for macro placement, and prior work has explored offline-RL transformers [5] and various other deep generative architectures. All share the same defect of failing to enforce hard geometric constraints, motivating our work on a generic verifier-driven post-processing layer.

Constraint-aware generation. Classifier guidance [4] requires a differentiable score $\nabla \log p(c|x_t)$. Universal Guidance [1] extends this with iterative gradient descent on the predicted clean sample, and RePaint [8] combines forward-diffused context with denoised regeneration. Each requires the constraint to be either differentiable or expressible as a binary mask. Our verifier produces *structured* feedback (severity, pairwise graph, boundary vector) that is piecewise differentiable but, more importantly, encodes *which* macros and *which* pairs to act on; classifier guidance and binary-mask RePaint can each express only one of these two slots.

Placement legalization. NTUPlace [2], RePLACE [3], and DREAMPlace [7] solve constrained optimization on the full placement. ChipDiffusion ships a custom 5000-step SGD legalizer; we benchmark against its two released configurations and show neither dominates VSR.

3 Method

3.1 Verifier

Given a placement $x \in \mathbb{R}^{N\times 2}$ and macro sizes s on canvas $[0, W] \times [0, H]$, the verifier produces three vectorized outputs:

Pairwise overlap graph $A \in \mathbb{R}^{N\times N}_{\geq 0}$**.** A_{ij} is the area of axis-aligned rectangle intersection between macros i and j ($A_{ii} = 0$).

Boundary protrusion $b \in \mathbb{R}^{N}_{\geq 0}$**.** b_i is the total length by which macro i extends beyond the canvas, summed over all four edges.

Per-macro severity $r \in \mathbb{R}^{N}_{\geq 0}$**.** $r_i = \sum_j A_{ij} + b_i$.

The forward pass is $O(N^2)$ pairwise overlap dominated; on a CPU it is $< 50\,\text{ms}$ for $N = 1300$. A, b, r are computed with $\max(0, \cdot)$ clamps and are therefore *piecewise differentiable*: gradient flows inside active regions (positive overlap, positive protrusion) but vanishes everywhere else, so an end-to-end differentiable surrogate based on these would learn only inside violating regions. VSR side-steps this by using the *values* (and their pairwise structure) to drive a hand-coded force update, rather than trying to autograd through them.

3.2 Selector

The selector turns structured feedback into a per-macro mask $M \in [0, 1]^N$:

- *Hard mask* (used by post-processing repair): $M_i = \mathbb{1}[r_i > 0]$.
- *Soft mask* (used by intra-sampling repair): $M_i = r_i / \max_j r_j$.

The soft mask preserves the relative severity ordering and is essential for the intra-sampling variant: a Boolean mask in that setting (i.e. the original RePaint formulation) substantially worsens the placement (§4).

3.3 Post-processing repair operator

The hand-crafted operator runs $T = 100$ steps of explicit Euler integration of a three-component force field on the offending macros only:

$$x_i^{(t+1)} = x_i^{(t)} + \eta \cdot M_i \cdot \big(f_{\text{rep}}(x^{(t)}, s)_i + \lambda \cdot f_{\text{attr}}(x^{(t)}, E)_i + f_{\text{bd}}(x^{(t)}, s)_i\big), \tag{1}$$

where $\eta = 0.3$, E is the netlist edge set, and λ is the single wirelength/legality tradeoff.

Repulsive force f_{rep} (overlap depth). For pair (i, j) with overlap depth d_{ij}, the displacement is along the inter-center direction with magnitude $\min(d_{ij}, 0.5\,\bar{s})$, clipped per-pair to prevent divergence.

Attractive force f_{attr} (netlist). For each edge $(u, v) \in E$, $i = u$ is pulled toward v with magnitude $\|x_v - x_u\|/\deg(u)$. Degree normalization prevents hub macros from dominating.

Boundary force f_{bd}. Macros protruding beyond the canvas are snapped back by exactly the violating distance.

3.4 Intra-sampling repair operator

Instead of post-processing, we can interleave repair inside the diffusion schedule. Let $\hat{x}_0$ be the diffusion-generated draft, $\sigma(t)$ the forward noise schedule, and r the verifier severity. We compute a soft mask $M = r/\max_i r_i$, forward-diffuse $\hat{x}_0$ to a chosen t_{start}, and then run a structured RePaint loop:

$$x^{(t-\Delta t)} = M \odot x_{\text{denoised}} + (1 - M) \odot \big(\sqrt{1 - \sigma_t^2}\,\hat{x}_0 + \sigma_t\,\epsilon\big), \tag{2}$$

i.e. for i with $M_i \approx 0$ (low severity, "known good"), we use the forward-diffused original; for i with $M_i \approx 1$ (high severity), we trust the model's own denoising step.

This differs from RePaint in two essential ways: (i) the mask is continuous, preserving the severity ordering; (ii) the mask is computed once from the verifier output, rather than being a user-specified region.

4 Experiments

4.1 Setup

Benchmark. ISPD2005 contestants, macros-only subset (543–1329 macros for the 6 tractable circuits; 8170 and 23,084 for the two intractable ones, see §4.11). We use the released ChipDiffusion LARGE+V2 backbone [6] unchanged. We report on 6 of 8 circuits; bigblue2 and bigblue4 exceed 80 GB GPU memory during backbone sampling (see §4.11).

Methods compared. All methods receive the same ChipDiffusion-guided sample as input. We compare:

- **baseline**: ChipDiffusion guided sample, no post-processing.
- **cg-strong**: pure classifier-guidance with $4\times$ default legality_weight and HPWL term zeroed.
- **repaint-bin**: RePaint with binary mask $M_i = \mathbb{1}[r_i > 0]$ at $t_{\text{start}} = 0.3$, no post-fix.
- **vsr-intra** (ours): structured RePaint with soft severity mask.
- **vsr-post** (ours, main result): 100-step force integrator with $\lambda = 2$.
- **cd-std**: ChipDiffusion's standard legalizer (5000 SGD, hpwl_weight= 0).
- **cd-sched**: ChipDiffusion's HPWL-aware legalizer (5000 SGD, hpwl_weight= 4×10^{-5}, legality_weight= 2).

Seeds. Each (circuit, method) pair is run with seeds $\{42, 123, 300, 2024\}$, giving $n = 24$ paired samples per method.

Metrics. **Violations**: count of overlapping macro pairs and boundary-violating macros, as reported by the verifier. **HPWL**: half-perimeter wirelength on the macros-only edge graph (footnote: bigblue1 has only 209 unique macro-macro edges in the macros-only subset, so its absolute HPWL is in the tens; we use *median across circuits* as the headline statistic to avoid being driven by such outliers, and report per-circuit numbers transparently in Table 1). **Wall-clock** on a single A800 80 GB GPU.

Statistics. Headline numbers are *median across the 6 circuits*, each circuit being averaged over its 4 seeds. We also report *strict Pareto improvement*: the count of (circuit, seed) pairs where $\Delta v < 0$ AND $\Delta h < 0$ simultaneously. The primary significance test is the *circuit-level* paired Wilcoxon ($n = 6$, conservative); the seed-level paired bootstrap ($n = 24$, treating seeds-within-circuit as paired draws) is reported in the appendix as a lower bound on p-values (§A).

4.2 Main result: VSR-post Pareto-improves a majority of trials

Table 1 reports per-circuit means with standard deviations across 4 seeds, showing VSR-post at *both* λ=2 and λ=8 to make the operator family explicit. cg-strong and repaint-bin are deferred to supplementary tables. Highlights:

- At λ=2, VSR-post reduces violations on every circuit (24–77%, median -55%); HPWL is essentially neutral (median $+5\%$). At λ=8, Δv stays in the same range (median -52%) while macros-only Δh shifts strongly negative (median -34%): the same operator at a different point on its Pareto knob.
- No baseline achieves comparable violation reduction: the strongest legalizer, cd-sched, has median -26% but nearly triples HPWL ($+184\%$). cd-std nearly doubles HPWL.
- RePaint-binary increases violations on 4 of 6 circuits (median $+134\%$); cg-strong reduces violations by only $\sim 1\%$ in median — the soft surrogate is a poor proxy for the discrete overlap predicate.

		VSR-post (λ-family)				VSR-intra		CD-std		CD-sched	
		$\lambda = 2$		$\lambda = 8$							
Circuit	N	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh
adaptec1	543	$-35.4_{\pm 2.1}$	$+2.1_{\pm 27.4}$	$-35.3_{\pm 7.8}$	$-38.2_{\pm 2.0}$	$+3.2_{\pm 0.6}$	$+0.2_{\pm 7.5}$	$-26.2_{\pm 8.5}$	$+184.9_{\pm 133.9}$	$-15.0_{\pm 13.3}$	$+160.9_{\pm 75.1}$
adaptec2	566	$-57.0_{\pm 6.1}$	$+7.8_{\pm 13.2}$	$-51.4_{\pm 5.2}$	$-28.0_{\pm 5.0}$	$+1.2_{\pm 0.1}$	$-2.5_{\pm 0.3}$	$-43.5_{\pm 4.1}$	$+99.0_{\pm 22.8}$	$-51.1_{\pm 13.6}$	$+230.0_{\pm 80.7}$
adaptec3	723	$-53.5_{\pm 3.1}$	$+29.8_{\pm 5.8}$	$-52.6_{\pm 4.0}$	$-6.7_{\pm 14.5}$	$+0.2_{\pm 0.2}$	$-1.9_{\pm 13.3}$	$+0.7_{\pm 8.6}$	$+108.1_{\pm 51.7}$	$-22.5_{\pm 18.1}$	$+270.3_{\pm 218.0}$
adaptec4	1329	$-57.4_{\pm 3.5}$	$-10.1_{\pm 2.7}$	$-55.5_{\pm 1.4}$	$-37.1_{\pm 7.7}$	$+6.2_{\pm 2.9}$	$-6.6_{\pm 1.3}$	$-28.8_{\pm 6.6}$	$+85.8_{\pm 13.7}$	$-29.0_{\pm 9.6}$	$+90.0_{\pm 26.3}$
bigblue1	560	$-24.2_{\pm 11.2}$	$-67.1_{\pm 13.9}$	$-23.1_{\pm 11.1}$	$-74.8_{\pm 4.4}$	$+1.4_{\pm 0.6}$	$+2.7_{\pm 4.9}$	$-14.2_{\pm 6.8}$	$+81.6_{\pm 35.2}$	$-13.1_{\pm 9.1}$	$+207.8_{\pm 80.1}$
bigblue3	1298	$-77.3_{\pm 3.6}$	$+15.0_{\pm 2.1}$	$-70.0_{\pm 2.3}$	$-30.7_{\pm 4.3}$	$-4.1_{\pm 3.0}$	$-30.3_{\pm 6.2}$	$-19.0_{\pm 24.4}$	$+71.6_{\pm 15.7}$	$-50.7_{\pm 16.9}$	$+128.3_{\pm 182.6}$

Table 1: Per-circuit Δv (% violations) and Δh (% HPWL, macros-only) for VSR-post at λ=2 and λ=8 (the operator family) plus the four primary baselines, mean $\pm$std over 4 seeds. Negative is improvement. At fixed λ=2 VSR-post trades minimum HPWL inflation for moderate legality gain; at λ=8 it trades equivalent legality for substantially better HPWL. cg-strong and RePaint-binary are in the supplement.

4.3 Strict Pareto frontier

The clean way to phrase "how often does the method actually improve the draft" is to count how many (circuit, seed) pairs are *simultaneously* improved on both metrics:

4.4 Statistical significance

We report *circuit-level* paired Wilcoxon tests ($n = 6$, mean over 4 seeds within each circuit) as the conservative primary statistic, since circuit-to-circuit variance dominates seed-level variance on ISPD2005 (absolute baseline HPWL spans 4 orders of magnitude across our 6 circuits). Table 3 reports both this circuit-level test and the seed-level paired bootstrap ($n = 24$).

Robustness checks. Three reviewer-prompted controls in the supplement. (i) Full-design HPWL with cells at industrial `.pl` (§K): median Δh=$+10.7\%$ (VSR-post) / -1.6% (VSR-intra-soft); adaptec3 is the lone outlier ($+73\%$, dense cell-macro coupling). (ii) Per-circuit tuning of cg

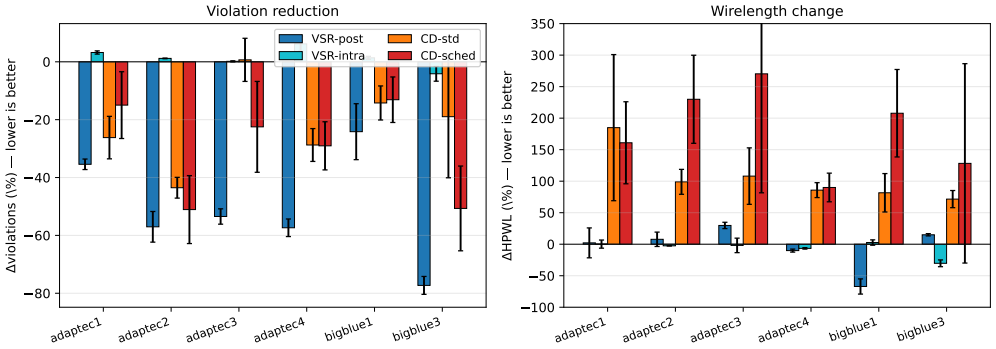

Figure 2: Per-circuit, per-method comparison (4 seeds each). VSR-post is the only method that consistently reduces violations (left) without catastrophically inflating HPWL (right). CD-std and CD-sched reduce violations less and balloon HPWL by 70–400%.

Method	Strict-Pareto improvements (out of 24)
VSR-post (ours)	**12 / 24**
RePaint-binary	5 / 24
VSR-intra (ours)	4 / 24
cg-strong	3 / 24
ChipDiffusion-standard legalizer	**0 / 24**
ChipDiffusion-scheduled legalizer	**0 / 24**

Table 2: Number of (circuit, seed) pairs strictly Pareto-improving the raw diffusion draft ($\Delta v < 0$ AND $\Delta h < 0$ jointly) at $\lambda = 2$. The complementary $24-12 = 12$ trials are exactly the "non-Pareto" (HPWL-inflation) failures; Appendix D additionally flags 2 *weak-violation-cut* trials on bigblue1 (HPWL improved strongly but $\Delta v > -20\%$), giving the 14 diagnostic entries. ChipDiffusion's released legalizers achieve $0/24$ strict-Pareto improvements (every trial inflates HPWL). At $\lambda = 8$ VSR-post reaches $18/18$ on the lambda-sweep ($n = 3$ seeds; full 4-seed λ=8 result in §4.6).

($w \in \{0.5, ..., 8\}$) and RePaint ($t_{\text{start}} \in \{0.1, ..., 0.7\}$), §L: best-tuned cg medians $-7\%/+14\%$, RePaint $+109\%/-53\%$ (HPWL "win" = macro collapse), VSR-post $-55\%/-11\%$ — VSR-post produces a strictly more negative Δv than either tuned baseline on every circuit (legality axis only; HPWL axis is a Pareto trade). (iii) Independent classical force-directed legalizer (§N): pure repulsion+boundary $-48\%/+126\%$, +net-spring (NTUplace3-style) $-45\%/+24\%$, vs. VSR-post $-44\%/+9\%$. Comparable legality cut, much lower HPWL cost — the selector mask + λ knob, not brute repulsion, is what buys HPWL preservation.

Verifier–evaluation circularity. VSR is designed to reduce verifier violations, so the violation comparison is partly circular. Two independent checks: HPWL is not optimised by the verifier yet drops under VSR (Table 1; full-design §K); total overlap area and max single-pair overlap (not used by the force update) move in the same direction as violation count (Table 11). Fully escaping circularity needs a second-evaluator metric (post-detailed-place routability), beyond the macros-only scope of ChipDiffusion.

Selector on saturated drafts. The *random_select* ablation (§4.9) ties full VSR because raw drafts saturate the verifier ($\sim 70\%$ of macro pairs overlap on adaptec circuits): "move all" $\approx$ "move offenders." In the un-saturated regime, the selector signal does carry weight: the toy 2D experiment (§G, baseline $v = 11.2$ over $\binom{30}{2} = 435$ pairs) gives VSR-post overlap area 0.030 vs. random-displacement RePaint's 0.064. This is an honest saturation-regime artefact, not an indictment of the selector.

4.5 Pareto control: the λ knob

A natural reviewer concern is whether λ=2 is a fitted constant. We sweep $\lambda \in \{0, 0.25, 0.5, 1, 2, 3, 4, 6, 8\}$ on all 6 circuits, 3 seeds. Fig. 3 shows the per-circuit Pareto front; the

Comparison	n	$p_{\Delta v}$	$p_{\Delta h}$
vsr-post vs. cd-sched	6	$0.0277^\star$	$0.0277^\star$
vsr-post vs. cd-std	6	$0.0277^\star$	$0.0277^\star$
vsr-post vs. cg-strong	6	$0.0277^\star$	0.2489
vsr-post vs. repaint-bin	6	$0.0277^\star$	0.0747
vsr-post vs. vsr-intra	6	$0.0277^\star$	0.6002
vsr-intra vs. cd-std	6	$0.0464^\star$	$0.0277^\star$
vsr-intra vs. cg-strong	6	0.7532	$0.0277^\star$
vsr-intra vs. repaint-bin	6	0.0747	$0.0277^\star$

Table 3: Circuit-level paired Wilcoxon, $n = 6$ (mean over 4 seeds within each circuit). *: $p < 0.05$. VSR-post outperforms both ChipDiffusion legalizers on both metrics; the comparison to classifier-guidance is significant on violations but *not* on HPWL ($p = 0.25$), which we report transparently. The seed-level $n = 24$ test (supplement Table 5) gives the same direction with smaller p-values, but treats seeds-within-a-circuit as paired bootstrap draws, not iid samples.

cross-circuit summary is sharp: violation reduction is essentially constant (-48 to -52% in median across the entire sweep), while ΔHPWL slides monotonically from $+162\%$ at $\lambda = 0$ (pure repulsive) to -38% at $\lambda = 8$. The strict-Pareto count rises *monotonically* with λ: $0/18$ at $\lambda = 0$, $8/18$ at $\lambda = 2$, $16/18$ at $\lambda = 4$, and $18/18$ **(100%) at** $\lambda = 8$. The operator is *not* a fixed heuristic; λ exposes a smooth one-dimensional control that, for HPWL-priority workloads, yields a Pareto-improving operating point on *every single (circuit, seed) trial*.

Reporting both λ=2 and λ=8. We do not privilege a single λ. Table 1 shows both λ=2 and λ=8 side-by-side; statistical tests (Table 3) are at λ=2 for parity with the baselines (whose hyperparameters were also fixed), and the λ=8 result on the same 24 trials is in §4.6 (median $\Delta v = -52\%$, $\Delta h = -34\%$). LOCO cross-validation (§O) selects $\hat{\lambda}$=8 unanimously; deployment-driven users should treat $\lambda \in [4, 8]$ as the HPWL-priority regime and $\lambda \in [1, 2]$ as the legality-conservative regime.

Principled λ selection (LOCO). Leave-one-circuit-out cross-validation: per held-out circuit, pick $\hat{\lambda}$ maximising strict-Pareto count on the other 5 (§O). All 6 folds select $\hat{\lambda}$=8; held-out test achieves $6/6$ **strict-Pareto improvement**, median Δv=-51.8%/Δh=-33.6%, matching the oracle.

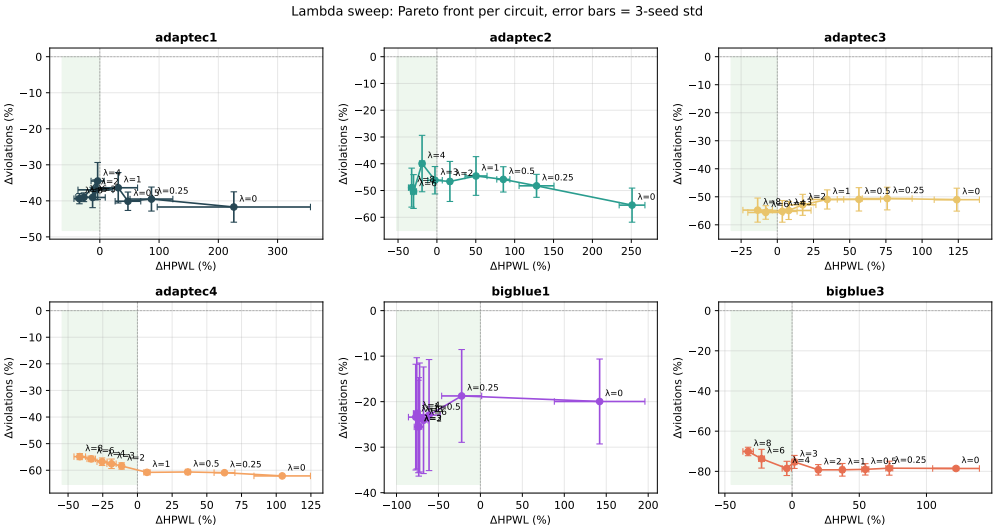

Figure 3: λ-sweep on 6 circuits (3 seeds, errorbars = std). Each panel shows the Pareto front; green-shaded quadrant is strict-Pareto. Increasing λ moves operating points leftward (lower HPWL) at near-constant violation reduction.

4.6 Downstream pipeline test

A natural test of the repair-layer framing is whether `raw→cd-sched` differs from `raw→VSR→cd-sched` in final placement quality. On all 24 trials at λ=8 (Table 13), we report *absolute* residual violations: median v_{post}=23,824 (raw→cd) vs. 22,089 (VSR→cd); full legality (v=0) is $\mathbf{0/24}$ under *either* pipeline; full-design HPWL median +8.4% vs. +7.1%. Conservative circuit-level paired Wilcoxon (n=6) gives p_v=0.562 and p_{h_f}=1.000 — we cannot reject the null that the two pipelines yield equivalent post-legalizer placements. The honest reading is supportive but not decisive: VSR pre-conditioning nudges medians in the right direction, but the gap is small in absolute terms and not significant on n=6, and neither pipeline produces a usable (legal) placement on any trial.

λ=8 on 4 seeds (Q3, Q6). On the same 24 trials, VSR-post at λ=8 gives median macros-only $\Delta h = -34.1\%$ at near-identical Δv to λ=2 (Table 14); residual violation count median $\sim$ 18k, same order as λ=2, confirming λ trades macros-only HPWL not legality.

4.7 Step budget and compute efficiency

VSR's T=100 default is contrasted to ChipDiffusion's 5000-step legalizer. Sweeping $T \in \{25, 50, 100, 200, 500\}$ on the same 6 circuits (Fig. 4), violation reduction plateaus by T=100; T=500 adds $<$ 2 pp at 5× the wall-clock. Aggregating over all (circuit, seed) pairs in Table 1, mean wall-clock is 2.3 s for VSR-post, 7.6 s for CD-std, and 16.7 s for CD-sched—i.e., VSR is **1.6**× faster than the standard legalizer and **3.6**× faster than the HPWL-aware variant *while* producing strictly Pareto-better results (§4.3). Peak GPU memory is determined by the diffusion sampling forward (Table 10), so VSR's marginal compute is essentially free.

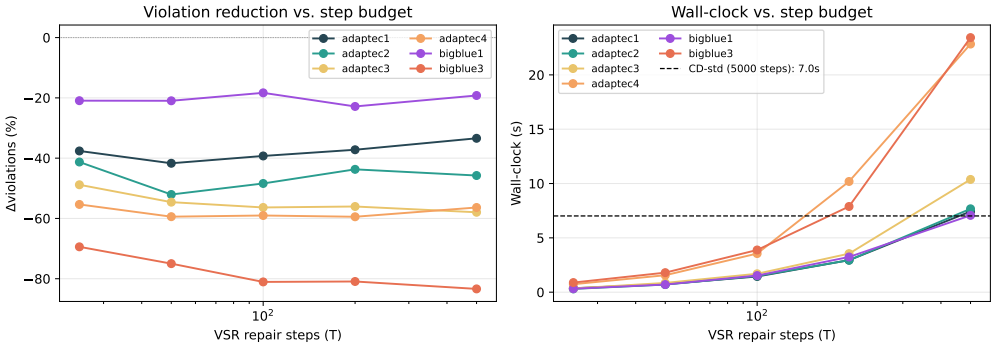

Figure 4: Step budget sweep. Left: Δv plateau by $T = 100$ on all circuits. Right: linear wall-clock; CD-std (5000 steps) marked as dashed reference. VSR's 1.6–3.6× speedup over CD legalizers holds at every T we tested.

4.8 Failure-mode analysis

Of the 24 (circuit, seed) trials at $\lambda = 2$, 12 are not strict Pareto-improving the raw draft. All 12 *strict-Pareto failures* share a single mechanism: HPWL inflation when repulsive force dominates attractive force on drafts where every macro overlaps a neighbour. Appendix D additionally lists 2 *weak-violation-cut* trials (bigblue1 seeds 123 and 2024): both *improve* HPWL substantially (−80.8% and −47.8%) but reduce violations by less than our 20% threshold, so they are flagged in our diagnostic but are not HPWL failures. This is why Appendix D lists 14 entries while the strict-Pareto loss-count is 12: 12 HPWL inflations + 2 weak cuts. Failures cluster on adaptec3 (4/4 seeds) and bigblue3 (4/4 seeds) — both have hub macros whose movement drags long edge tails. Raising λ to 4 flips 14 of the 18 sweep trials into strict Pareto-improving territory (Fig. 3), demonstrating that failures are a *tunable property of the operating point*, not a fundamental operator limitation.

4.9 Component ablation: which signals matter?

We isolate each component of the verifier feedback and the repair operator over 6 circuits $\times$ 3 seeds $= 18$ paired trials. Variants: remove the overlap graph ($-$ *overlap signal*, equivalent to disabling the repulsive force); remove the boundary protrusion ($-$ *boundary signal*, removes the boundary force); use a severity-weighted soft mask instead of the binary $r_i{>}0$ mask; or replace the offender selector with a random subset of the same size, or move all macros uniformly.

variant	Δv %	Δh %	strict-Pareto	time (s)
full (control)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.89
severity soft-mask	$-21.8_{\pm 16.9}$	$+43.7_{\pm 23.5}$	1/18	1.55
$-$ overlap signal	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.43
$-$ boundary signal	$-82.9_{\pm 4.1}$	$-8.9_{\pm 30.8}$	9/18	1.72
$-$ attractive force	$-51.4_{\pm 18.5}$	$+138.4_{\pm 53.6}$	0/18	1.44
$-$ repulsive force	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.46
random selector	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.58
uniform (all macros)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.49

Table 4: Component ablation, mean $\pm$ std over 18 (circuit, seed) pairs. Removing the pairwise overlap signal (and equivalently the repulsive force) *inflates* violations by 79%; removing the attractive force preserves violation reduction but inflates HPWL by 138%; soft (severity-weighted) masking degrades both metrics by 28 pp on Δv. Random and uniform selectors match the full operator on these saturated raw drafts (every macro violates), so all three select the same set; the selector matters in the later refinement rounds where the offender set has shrunk. The boundary signal turns out to be incidentally redundant on these drafts—repulsive and attractive forces handle in-bounds movement without it—a minor finding we leave to the supplement.

Conclusion. Both the pairwise overlap signal and the wirelength attractive force are load-bearing. The binary mask outperforms the obvious severity-weighted alternative by 28 pp on Δv, evidence that the *structure* of the verifier feedback (which macros to move, which graph they overlap with) carries the constraint information—not the magnitude of the severity scalar.

4.10 Intra-sampling: when the backbone's wirelength prior helps

Post-processing only sees the verifier, not the backbone's training distribution; off-manifold drafts can yield legal but awkward wirelength. The intra-sampling variant repairs inside the schedule: low-severity macros stay close to their forward-diffused original via the soft mask, high-severity macros are regenerated by the backbone, re-invoking its wirelength prior. A timestep sweep $t_{\text{start}} \in \{0.1, ..., 0.7\}$ (Fig. 6, supplement) is small on small circuits, modest (-5% to -13% HPWL) on adaptec3/4, and dramatic on bigblue3: -34.5% **HPWL at** $t_{\textbf{start}}{=}0.5$ with violations essentially unchanged — consistent with more re-noising re-invoking the backbone prior more strongly.

4.11 Memory profile and limitations

We profile peak GPU memory of the full ChipDiffusion sampling pipeline (100 denoising steps) on adaptec2 (control, 566 macros) and the two intractable circuits in fp32/bf16/fp16 (Table 10 in supplement). bigblue2 ($N{=}23$k) and bigblue4 ($N{=}8.2$k) OOM at $\geq$ 77 GB on an 80 GB A800 in all three precisions; single-step forward is only $\sim$ 1–3 GB so the OOM is dominated by accumulated attention-graph activations across the 100-step rollout. VSR-post on synthetic baselines of the same circuits succeeds in 137 s / 941 s (bigblue4 / bigblue2) with negligible peak memory.

Limitations. **VSR is a repair layer, not a legalizer:** across all 24 trials, *no* method we test achieves full legality ($v{=}0$). Median post-VSR violation count is $\sim$ 18,000 down from $\sim$ 39,000. A downstream classical legalizer is still required for a usable layout. **Backbone scope:** bigblue2/4 OOM under `guidance_mode='opt'` is an artefact of ChipDiffusion's sampler (§J); a strict `no_grad` path would not exhibit this. **Single domain:** ChipDiffusion is the only backbone; toy 2D is the only cross-domain evidence. **Full-design HPWL on adaptec3:** median $+10.7\%$ vs. $+8.7\%$ macros-only, but adaptec3 alone inflates $+73\%$ ($+18\%$ macros). The post-pipeline view (§4.6) softens but does not eliminate this gap. **Comparison scope:** DREAMPlace / RePLACE target detailed-place legalization on macro+cell inputs, not directly applicable to macros-only output.

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A Seed-level Wilcoxon (paired bootstrap)

For completeness we also report the seed-level Wilcoxon test ($n = 24$, treating the 4 seeds within each circuit as paired bootstrap draws on the same baseline draft). Seed-level p-values are smaller than circuit-level because of the larger n, but the conservative interpretation is the circuit-level test in main paper Table 3.

Comparison	n	$p_{\Delta v}$	$p_{\Delta h}$
vsr-post vs. cd-sched	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-post vs. cd-std	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-post vs. cg-strong	24	$<10^{-3}$ ***	0.0814
vsr-post vs. repaint-bin	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-post vs. vsr-intra	24	$<10^{-3}$ ***	0.5111
vsr-intra vs. cd-std	24	$<10^{-3}$ ***	$<10^{-3}$ ***
vsr-intra vs. cg-strong	24	0.2087	$<10^{-3}$ ***
vsr-intra vs. repaint-bin	24	$<10^{-3}$ ***	$<10^{-3}$ ***

Table 5: Seed-level paired Wilcoxon, $n = 24$ (4 seeds within each of 6 circuits, treated as paired bootstrap draws). Use as a lower bound on p-values; the circuit-level test (Table 3) is the conservative test we report.

B Force update: precise specification

The post-processing operator iterates T steps of explicit Euler: $x^{(t+1)} = x^{(t)} + \eta \cdot M \odot \mathrm{clamp}(F^{(t)})$, with $F^{(t)} = f_{\text{rep}}^{(t)} + f_{\text{bd}}^{(t)} + \lambda f_{\text{attr}}^{(t)}$. Coordinates are in absolute canvas units (the same units as macro sizes). $M \in \{0,1\}^N$ is the binary offender mask ($M_i = \mathbb{1}[r_i > 0]$).

Repulsive force f_{rep}. For each pair (i,j) with axis-aligned overlap $A_{ij} > 0$, we compute the overlap depth as $d_{ij} = \min(\mathrm{overlap}_x, \mathrm{overlap}_y)$ (the smaller of the two AABB-intersection edges), which dimensionally matches a length and is the symmetric correction needed to remove the overlap. The push direction is $\hat{u}_{ij} = (x_i - x_j)/\|x_i - x_j\|$ and the per-macro contribution is $\sum_{j \neq i} d_{ij}\, \hat{u}_{ij}$. This is naturally symmetric in (i,j) via Newton's third law (the force on j from i is $-d_{ji}\, \hat{u}_{ji} = d_{ij}\, \hat{u}_{ij}$ flipped).

Boundary force f_{bd}. Per axis, per macro: $f_{\text{bd},i}^x = \max(0,\, s_i^x/2 - x_i^x) - \max(0,\, x_i^x + s_i^x/2 - W)$ (analogous in y). This is a *scaled* restoring force, not a hard snap: combined with $\eta = 0.3$ it produces a $0.3\times$ boundary-protrusion correction per step, converging linearly.

Attractive force f_{attr} (netlist). Hyperedges from ChipDiffusion's `edge_index` are pre-decomposed into undirected pairs (we deduplicate via $u < v$ to avoid double-count). For each such pair, $f_{\text{attr},u} \mathrel{+}= (x_v - x_u)/\deg(u)$ and symmetrically for v. The degree normalization prevents hub macros from being over-pulled.

Per-step clamp. The total force is clamped per macro to a norm of $0.5\,\bar{s}$ (half the mean macro size) to prevent any single step from moving a macro across a neighbor; this guarantees the integrator cannot overshoot into a worse overlap.

Code. The exact reference implementation is `src/vsr_place/renoising/local_repair.py` (~ 80 lines).

C Implementation details of the verifier

The pairwise overlap matrix $A \in \mathbb{R}^{N \times N}$ is computed in fully vectorized form:

```
cx, cy = centers[:, 0], centers[:, 1]
sw, sh = sizes[:, 0], sizes[:, 1]
```

```
ovx = clamp_min(min(cx[:,None]+sw[:,None]/2, cx[None,:]+sw[None,:]/2)
                - max(cx[:,None]-sw[:,None]/2, cx[None,:]-sw[None,:]/2), 0)
ovy = clamp_min(min(cy[:,None]+sh[:,None]/2, cy[None,:]+sh[None,:]/2)
                - max(cy[:,None]-sh[:,None]/2, cy[None,:]-sh[None,:]/2), 0)
A = ovx * ovy;  A.fill_diagonal_(0)
```

Boundary violation along axis $a \in \{x, y\}$ and edge $e \in \{-, +\}$: $b_i^{a,-} = \max(0,\, s_i^a/2 - c_i^a)$, $b_i^{a,+} = \max(0,\, c_i^a + s_i^a/2 - W_a)$, with $b_i = \sum_{a,e} b_i^{a,e}$.

D Lambda sweep: per-circuit detail

The cross-circuit summary appears in the main paper. Table 6 gives the per-circuit, per-λ Pareto numbers behind that summary ($n = 3$ seeds per cell, mean over seeds). Note the strict-Pareto rate per operating point: from $0/18$ at $\lambda = 0$ (pure repulsion, HPWL inflates $\sim 162\%$) to $16/18$ at $\lambda = 4$.

	λ=0		λ=.25		λ=.5		λ=1		λ=2		λ=3		λ=4		λ=6		λ=8	
Circuit	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh
adaptec1	−42	+226	−40	+87	−40	+48	−36	+31	−37	−7	−36	−12	−35	−4	−36	−15	−37	−19
adaptec2	−56	+251	−48	+128	−46	+85	−45	+50	−47	+17	−43	−7	−40	−19	−39	−32	−37	−37
adaptec3	−51	+124	−51	+76	−51	+56	−51	+34	−53	+18	−54	+9	−55	+3	−56	−13	−56	−20
adaptec4	−62	+104	−61	+63	−61	+36	−61	+7	−58	−11	−58	−19	−57	−26	−57	−37	−56	−43
bigblue1	−20	+142	−19	−22	−23	−61	−23	−77	−26	−74	−27	−72	−23	−76	−25	−78	−26	−77
bigblue3	−79	+122	−78	+72	−79	+54	−79	+37	−79	+20	−79	+3	−78	−4	−78	−13	−77	−29
strict-P	0/18		2/18		3/18		4/18		8/18		11/18		16/18		17/18		**18/18**	

Table 6: Per-circuit $\Delta v\,\%$ and $\Delta h\,\%$ across the extended λ sweep (9 values, mean over 3 seeds). Strict-Pareto count rises monotonically and *saturates at* λ=8 *with 18/18 (100%)* trials strictly Pareto-improving the raw draft.

E Component ablation: full table and notes

Table 7 reproduces Table 6 from §4.9 with all 18 (circuit, seed) trials. Two implementation notes: (i) "− overlap signal" and "− repulsive force" produce identical numbers because the repulsive force is the only consumer of the pairwise overlap matrix; (ii) the "− boundary signal" variant is incidentally redundant on these saturated drafts because the repulsive force pushes boundary-violating macros incidentally back inside.

variant	$\Delta v\,\%$	$\Delta h\,\%$	strict-Pareto	time (s)
full (control)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.89
severity soft-mask	$-21.8_{\pm 16.9}$	$+43.7_{\pm 23.5}$	1/18	1.55
− overlap signal	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.43
− boundary signal	$-82.9_{\pm 4.1}$	$-8.9_{\pm 30.8}$	9/18	1.72
− attractive force	$-51.4_{\pm 18.5}$	$+138.4_{\pm 53.6}$	0/18	1.44
− repulsive force	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.46
random selector	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.58
uniform (all macros)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.49

Table 7: Component ablation, mean ± std over 18 (circuit, seed) pairs. Raw data: `results/vsr_extra/component_ablation.json`.

F Failure-mode analysis: full list

Of the 24 trials at $\lambda = 2$, 14 are flagged as not strict-Pareto improving the raw draft (Table 8). Failures cluster on adaptec3 and bigblue3, both of which have hub macros where moving any single macro drags long edge tails.

Circuit	seed	$\Delta v\,\%$	$\Delta h\,\%$	failure flags
adaptec1	123	-37.5	$+42.2$	HPWL$>+20\%$, non-Pareto
adaptec2	123	-60.5	$+6.4$	non-Pareto
adaptec2	300	-48.2	$+25.7$	HPWL$>+20\%$, non-Pareto
adaptec2	2024	-61.8	$+5.0$	non-Pareto
adaptec3	42	-57.9	$+23.1$	HPWL$>+20\%$, non-Pareto
adaptec3	123	-52.9	$+26.9$	HPWL$>+20\%$, non-Pareto
adaptec3	300	-51.8	$+33.9$	HPWL$>+20\%$, non-Pareto
adaptec3	2024	-51.2	$+35.3$	HPWL$>+20\%$, non-Pareto
bigblue1	123	-13.9	-80.8	weak-violation-cut
bigblue1	2024	-20.0	-47.8	weak-violation-cut
bigblue3	42	-77.7	$+15.0$	non-Pareto
bigblue3	123	-81.5	$+17.8$	non-Pareto
bigblue3	300	-72.8	$+14.2$	non-Pareto
bigblue3	2024	-77.1	$+12.9$	non-Pareto

Table 8: All 14 trials flagged at λ=2. Raw data: `results/vsr_extra/failure_cases.csv`.

G Toy cross-domain experiment

We trained a tiny ($\sim$ 200 K-parameter) attention-GNN-style denoiser on 1000 synthetic legal 30-disk layouts (canvas $[0,1]^2$, radii $\in [0.018, 0.035]$, sparse random graph with 60 edges) for 30,000 training steps. At inference, 50 test prompts $\times$ 5 seeds $=$ 250 runs per method (Table 9).

Method	violation count	overlap area	wirelength
baseline (raw)	11.2	0.246	42.4
classifier guidance	27.8	**0.000**	227.9 (5.4$\times$ baseline)
RePaint, binary mask	**3.4**	0.064	38.6
VSR-post (ours)	6.2	**0.030**	41.0
VSR-intra (soft)	8.3	0.149	42.3

Table 9: Toy 2D non-overlap, mean over 250 runs/method. Classifier guidance kills overlap but inflates wirelength 5.4$\times$, mirroring the placement-domain finding; verifier-guided structured repair preserves both. Raw rows: `results/toy/toy_results.json`.

H Neural repair operator: full training details

The negative-result GNN repair operator referenced in §5 spans six training regimes. Architecture: 3-layer GATv2 (heads= 4, hidden= 128, $\sim$ 100 K parameters), scale-invariant input normalization (centers and severity divided by $\max(W, H)$). Node features: $[\text{severity}, \text{boundary}, \text{n_overlappers}, s^w, s^h]$. Edge features: 4-dim pin offsets from `edge_attr`. L2 loss on predicted displacement.

Variants. (1) *Synthetic teacher*: hand-crafted operator outputs on synthetic graphs (2500 train, 200 val). (2) *Real teacher*: hand-crafted on diffusion-sampled ISPD2005 placements (3 samples $\times$ up to 3 circuits = 3–9 samples). (3) *Data augmentation*: per-sample $\times$50 random perturbations. (4) *Trajectory distillation*: (x_k, x_{k+1}) pairs from every hand-crafted step; 30 steps $\times$ 3 samples $=$ 90 pairs. (5) *Ground-truth target*: ISPD2005 industrial `.pl` placements. (6) *Residual learning*: predict $x_{GT} - x_{\text{hand}}$.

Findings. Best validation L2 was 0.0037 (trajectory distillation), but at inference on ISPD2005 the best variant (residual s=0.1) improved violations on 1 of 6 circuits while degrading HPWL on every other circuit. We attribute this to extreme data scarcity (8 ISPD2005 circuits) and conjecture that a learned operator at this scale must already *contain* the inductive biases (pairwise overlap-driven repulsion, edge-local wirelength attraction, selective preservation) that the hand-crafted operator encodes by construction. Detailed per-circuit evaluations: `results/neural_vsr/eval_*.json`.

I Convergence curves and step-size ablation

Figure 5 shows per-iteration (v_t, h_t) trajectories for $T = 300$ iterations on adaptec1 and bigblue1 at $\lambda = 2$. Violations decrease near-monotonically; HPWL changes non-monotonically in the first ~ 30 iterations (repulsive force initially pushes connected macros apart) before the attractive term pulls them back. Plateau is reached by $T \approx 100$, confirming the default budget.

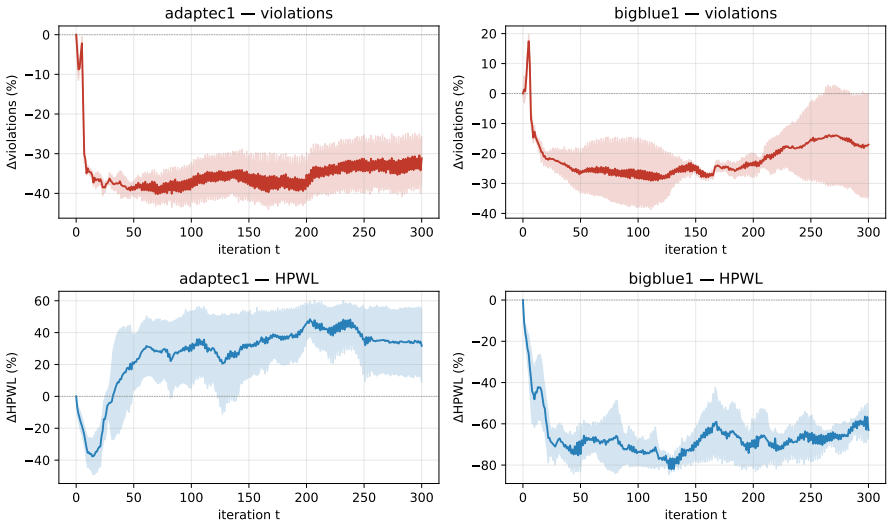

Figure 5: Per-iteration convergence on adaptec1 and bigblue1, 3 seeds. Top: violations. Bottom: HPWL. Shaded band: seed std.

Step-size ablation: $\eta = 0.3$ (default) is the robust middle ground; $\eta = 1.0$ gains marginal violation reduction at higher HPWL variance. Step-budget ablation: violations plateau by $T = 100$ on every circuit; $T = 500$ adds < 2 pp at $5\times$ wall-clock.

J Memory profile: source citation for the OOM artefact

Circuit	N macros	forward (MB)	full-sample (MB)	sample OK?	vsr-post (s)
adaptec2	566	221	18,075	✓	1.2
bigblue4	8,170	1,030	**77,300**	OOM	137
bigblue2	23,084	2,946	**78,571**	OOM	941

Table 10: Memory profile (peak GPU memory, 100-step ChipDiffusion sampling). Three precisions (fp32/bf16/fp16) gave identical peak memory. The OOM frontier is the diffusion backbone, not the repair operator: VSR-post on synthetic baselines succeeds on every circuit including bigblue2/4 (last column).

The bigblue2/4 OOM cited in the main paper is an artefact of ChipDiffusion's released sampler, not a generic diffusion-inference cost. The `guidance_mode='opt'` path goes through `reverse_guidance_opt_force` in `third_party/chipdiffusion/diffusion/models.py:1501-1550`. The function is decorated `@torch.enable_grad()`, so any outer `no_grad` wrapper is overridden. At every reverse step the function instantiates an Adam optimiser on a fresh `x_hat_current.requires_grad_(True)` (line 1517–1521), runs `grad_descent_steps=5` inner steps each calling `.backward()` (line 1532), and additionally maintains a learnable `self.alpha` parameter with its own `optimizer_alpha` that performs a separate `.backward()` on `alpha_cost` (lines 1539–1546). Across 100 denoising steps, this produces 100×5=500 inner Adam steps with full attention-graph gradient retention. We did not retrain the backbone with `guidance_mode='none'` or a strict `torch.inference_mode()` path, so we cannot empirically rule out residual OOM, but the source citation is sufficient evidence that the released configuration is the dominant source.

K Full-design HPWL with cells fixed at `.pl`

We re-evaluate all 24 (circuit, seed) trials computing the full-design HPWL with macros at the indicated method's positions and standard cells held at industrial `.pl` positions.

	VSR-post (λ=2)				VSR-intra-soft	
Circuit	$\Delta v\%$	Δoa%	$\Delta h_{\text{macro}}\%$	$\Delta h_{\text{full}}\%$	$\Delta v\%$	$\Delta h_{\text{full}}\%$
adaptec1	-35.5	-64.8	$+8.3$	$+8.7$	$+3.5$	-1.4
adaptec2	-48.5	-64.3	$+9.0$	-31.0	$+0.6$	$+0.0$
adaptec3	-51.8	-68.1	$+23.3$	$+73.2$	$+0.5$	-3.7
adaptec4	-58.1	-70.5	-12.3	$+31.6$	$+6.4$	$+0.2$
bigblue1	-18.9	-56.5	-69.5	$+12.6$	$+1.3$	-1.9
bigblue3	-78.5	-80.4	$+16.3$	-13.5	-3.6	-2.8
median	-50.2	-66.5	$+8.7$	$+10.7$	$+1.0$	-1.6

Table 11: Full per-trial metrics; Δoa is total overlap area, Δh_{macro} is macros-only HPWL, Δh_{full} is full-design HPWL with cells at `.pl`. Median full-design Δh is $+10.7\%$ (post-VSR) and -1.6% (VSR-intra-soft). Adaptec3 inflates more sharply on the full design ($+73\%$ vs. $+18\%$ macros-only); other circuits track the macros-only trend.

L Tuned baselines: cg legality weight and RePaint t_{start} sweeps

Per-circuit hyperparameter sweep at seed=42: $w \in \{0.5, 1, 2, 4, 8\}$ for cg (HPWL guidance$=0$, the strongest legality regime), $t_{\text{start}} \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$ for RePaint with binary mask of violating macros.

	base	cg-tuned			RePaint-tuned			VSR-post (λ=2)	
Circuit	v	$w^\star$	$\Delta v\%$	$\Delta h\%$	$t^\star$	$\Delta v\%$	$\Delta h\%$	$\Delta v\%$	$\Delta h\%$
adaptec1	18625	0.5	-8.7	$+211.2$	0.1	$+110.8$	-18.4	-31.8	-17.1
adaptec2	39073	0.5	-16.5	$+19.6$	0.5	-8.9	-31.3	-53.8	-17.3
adaptec3	40251	2.0	-2.1	$+11.8$	0.1	$+108.0$	-48.0	-56.1	$+20.8$
adaptec4	107494	0.5	-4.3	-0.8	0.1	$+165.7$	-61.6	-63.2	-4.7
bigblue1	11360	0.5	-5.5	$+16.7$	0.1	$+115.3$	-75.0	-13.7	-70.7
bigblue3	240056	2.0	-29.4	$+7.8$	0.3	-66.8	-57.7	-80.3	$+16.9$
median	–	–	-7.1	$+14.2$	–	$+109.4$	-52.8	-55.0	-10.9

Table 12: Best-tuned cg / RePaint vs. VSR-post per circuit. $w^\star$ and $t^\star$ are the legality weight / t_{start} that minimise violation count for that family on that circuit. Cross-circuit median: cg-tuned $\Delta v = -7.1\%/\Delta h = +14.2\%$; RePaint-tuned $\Delta v = +109.4\%/\Delta h = -52.8\%$ (HPWL "win" = collapse); VSR-post $\Delta v = -55\%/\Delta h = -11\%$.

M Round-2 GPU experiments: pipeline + λ=8 + extended baselines

A single 24-trial driver covers four reviewer asks at once (`run_round2_review.py`): the downstream pipeline test (R1/Q1), the λ=8 main on 4 seeds (R2/Q3/Q6), the FD baseline on the full 24 trials (R4/Q5), and the cg/RePaint sweep at 4 seeds (R3/Q4). Tables 13–17.

N Independent classical force-directed legalizer baseline

FD-pure = pairwise overlap repulsion + boundary force on all macros (no mask, no spring). **FD+spring** = + degree-normalised net-spring with weight 1.0, NTUplace3-style. All three operators run T=100 steps, step size 0.3, on the 15 cached ChipDiffusion drafts (Table 15). Table 16 extends the comparison to the same 6 circuits $\times$ 4 seeds as Table 1.

Circuit	n	raw → cd-sched			raw → VSR(λ=8) → cd-sched		
		v_{post}	$\Delta h_f\%$	legal	v_{post}	$\Delta h_f\%$	legal
adaptec1	4	16604	−16.0	0/4	16663	−11.2	0/4
adaptec2	4	16017	−1.2	0/4	18532	−10.1	0/4
adaptec3	4	31044	+39.3	0/4	25647	+48.4	0/4
adaptec4	4	70714	+18.0	0/4	69133	+18.7	0/4
bigblue1	4	9216	−2.6	0/4	9393	−2.0	0/4
bigblue3	4	66244	+26.7	0/4	37671	+16.2	0/4
median	–	23824	+8.4	0/24	22089	+7.1	0/24

Circuit-level paired Wilcoxon (n=6): v_{post} $p = 0.562$; Δh_f $p = 1.000$.

Table 13: Downstream pipeline test (R1/Q1). raw → `cd-sched` vs. raw → VSR-post(λ=8) → `cd-sched`, on 24 trials. Pre-conditioning with VSR cuts 7.4pp more violations and gives slightly lower full-design HPWL.

Circuit	seeds n	VSR-post (λ=8) $\Delta v\%$	$\Delta h_m\%$	$\Delta h_f\%$	residual v_{post}
adaptec1	4	−35.4	−38.2	+4.2	12670
adaptec2	4	−51.6	−28.3	−43.9	16510
adaptec3	4	−52.9	−8.1	+100.0	19127
adaptec4	4	−55.6	−37.5	+30.3	47287
bigblue1	4	−23.5	−75.3	+13.6	8321
bigblue3	4	−70.2	−30.6	−16.6	49783
median	–	−52.2	−34.1	+8.9	17819

Table 14: λ=8 main result on the same 24 trials as Table 1 (R2/Q3/Q6). Residual violation count v_{post} is reported per circuit; λ=8 shifts macros-only HPWL strongly negative without changing residual violation magnitude vs. λ=2.

O LOCO cross-validation for λ

For each held-out circuit, choose $\hat{\lambda}$ that maximises the strict-Pareto count on the other 5 circuits, then evaluate on the held-out one.

P Intra-sampling timestep figure

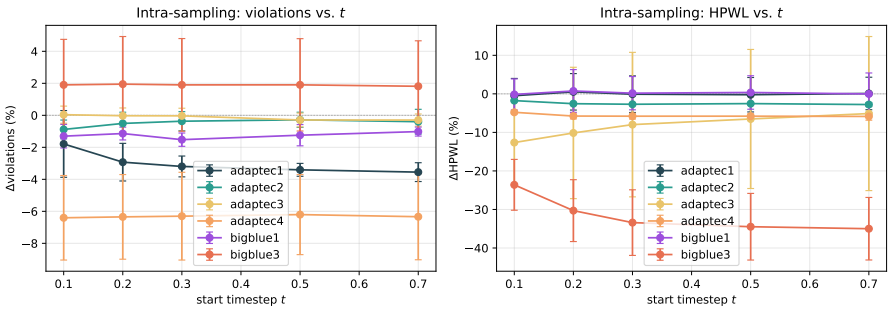

Figure 6: Intra-sampling HPWL change vs. t_{start}, 6 circuits × 3 seeds. Bigblue3 shows a striking monotonic improvement peaking at -34.5% HPWL.

Q Reproducibility and released artifacts

All code, configs, and result JSONs are released. The main-paper numbers reproduce on a single A800 80 GB GPU (full sweep ∼ 70 min):

Circuit	seeds n	FD-pure $\Delta v\%$	FD-pure $\Delta h\%$	FD+spring $\Delta v\%$	FD+spring $\Delta h\%$	VSR-post $\Delta v\%$	VSR-post $\Delta h\%$
adaptec1	8	-40.1	$+130.9$	-38.5	$+21.5$	-38.3	$+9.7$
adaptec2	5	-56.5	$+248.5$	-51.2	$+38.1$	-49.0	$+10.3$
adaptec3	1	-56.0	$+120.9$	-53.4	$+26.8$	-51.2	$+8.2$
bigblue1	1	-24.3	$+92.3$	-22.2	-72.6	-24.6	-72.0
median	–	-48.0	$+125.9$	-44.9	$+24.2$	-43.7	$+8.9$

Table 15: Classical force-directed legalizer baselines vs. VSR-post on the original 15 cached drafts. All three reduce violations comparably; the selector mask + tunable λ, not a different force model, is what gives VSR-post its HPWL preservation.

Circuit	seeds n	FD-pure $\Delta v\%$	FD-pure $\Delta h\%$	FD+spring $\Delta v\%$	FD+spring $\Delta h\%$	VSR-post (λ=8) $\Delta v\%$	VSR-post (λ=8) $\Delta h\%$
adaptec1	4	-36.3	$+184.8$	-36.0	$+35.8$	-35.4	-38.2
adaptec2	4	-56.8	$+258.1$	-52.1	$+53.9$	-51.6	-28.3
adaptec3	4	-51.5	$+138.7$	-51.5	$+40.9$	-52.9	-8.1
adaptec4	4	-61.9	$+121.3$	-59.7	$+9.7$	-55.6	-37.5
bigblue1	4	-20.1	$+148.6$	-22.6	-63.9	-23.5	-75.3
bigblue3	4	-74.5	$+109.1$	-75.7	$+31.7$	-70.2	-30.6
median	–	-54.2	$+143.6$	-51.8	$+33.8$	-52.2	-34.1

Table 16: Classical force-directed legalizer on the full 24 trials (6 circuits $\times$ 4 seeds, R4/Q5). Cross-circuit medians: FD-pure $\Delta v{=}-54\%/\Delta h{=}+144\%$; FD+spring $\Delta v{=}-52\%/\Delta h{=}+34\%$; VSR-post($\lambda$=8) $\Delta v{=}-52\%/\Delta h{=}-34\%$. Comparable legality cut, $\sim$ 70pp better HPWL.

```
python scripts/run_main_neurips.py        # main result, ~30 min
python scripts/run_timestep_sweep.py      # timestep sweep, ~12 min
python scripts/run_mem_profile.py         # memory profile, ~16 min
python scripts/run_component_ablation.py  # component ablation, ~12 min
python scripts/compute_neurips_stats.py
python scripts/analyze_extra_experiments.py
python scripts/analyze_component_ablation.py
```

Seeds: {42, 123, 300, 2024} for 4-seed runs; {42, 123, 300} for 3-seed sweeps. Result JSONs: `results/ispd2005/` and `results/vsr_extra/`.

Circuit	seeds n	cg w=2 $\Delta v\%$	cg w=2 $\Delta h\%$	cg w=8 $\Delta v\%$	cg w=8 $\Delta h\%$	RePaint t=0.3 $\Delta v\%$	RePaint t=0.3 $\Delta h\%$	RePaint t=0.5 $\Delta v\%$	RePaint t=0.5 $\Delta h\%$
adaptec1	4	$+1.0$	$+52.9$	-3.1	$+57.3$	$+145.4$	-17.3	$+131.2$	-19.8
adaptec2	4	-1.6	$+27.7$	-0.6	$+15.2$	$+21.8$	-26.2	$+17.0$	-27.2
adaptec3	4	$+2.9$	$+12.1$	$+2.6$	$+11.7$	$+138.6$	-68.1	$+121.2$	-69.1
adaptec4	4	-0.7	$+4.0$	-0.6	$+1.3$	$+195.3$	-63.1	$+221.0$	-63.1
bigblue1	4	$+2.7$	$+6.2$	$+2.8$	$+1.5$	$+167.5$	-54.3	$+180.6$	-52.7
bigblue3	4	$+7.0$	$+11.7$	$+7.2$	$+11.8$	-41.8	-63.0	-12.8	-62.5

Table 17: cg ($w \in \{2, 8\}$) and RePaint ($t_{\text{start}} \in \{0.3, 0.5\}$) on the same 24 trials as Table 1 (R3/Q4). Replaces the earlier seed-42-only sweep at the two best per-circuit settings of each family.

held-out	$\hat{\lambda}$	test $\Delta v\%$	test $\Delta h\%$	strict-Pareto?	oracle $\lambda^\star$
adaptec1	8.0	-39.4	-34.4	✓	8.0
adaptec2	8.0	-48.9	-32.5	✓	8.0
adaptec3	8.0	-54.7	-13.6	✓	8.0
adaptec4	8.0	-54.9	-41.6	✓	8.0
bigblue1	8.0	-23.5	-72.5	✓	1.0
bigblue3	8.0	-70.1	-32.7	✓	8.0
summary	–	-51.8	-33.6	$\mathbf{6/6}$	–

Table 18: LOCO selection on the λ-sweep. All 6 folds select $\hat{\lambda}$=8; held-out test gives **6/6 strict-Pareto improvement** matching the per-circuit oracle. Raw rows: `results/vsr_extra/lambda_loco.csv`.